



OPEN NanoTag is an improved method to map interactions between DNA and proteins not requiring IgG

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Genome-wide profiling of DNA–protein interactions in cells can provide important information about mechanisms of gene regulation. Most current methods for genome-wide profiling of DNA-bound proteins such as ChIP-seq and CUT&Tag use conventional IgG antibodies to bind the target protein(s). This limits their applicability to targets with available high affinity and specificity antibodies and prevents their use for other targets. Here we describe NanoTag, an IgG-free method derived from CUT&Tag to profile DNA–protein interactions. NanoTag is based on a fusion between an anti-GFP nanobody and Tn5 transposase that can map GFP-tagged proteins associated with chromatin in a fast, cost-effective and animal-free manner. We used NanoTag to indirectly profile the histone mark H3K4me3 genome-wide via its binding partner TATA box-binding protein-associated factor 3 (TAF3) and the transcription factors Nanog and CTCF in mouse embryonic stem cells (mESCs). NanoTag results show high inter-replicate reproducibility, high signal-to-noise ratio and strong correlation with CUT&Tag datasets, validating its accuracy and reliability. NanoTag provides a novel, flexible and cost-effective IgG-free method to generate high resolution DNA-binding profiles in cells and tissues.

Keywords Chromatin, DNA–protein interactions, Nanobody, CUT&Tag, Epigenomics, Tn5 tagmentation, Chromatin profiling

Abbreviations

ChIP-seq	Chromatin immunoprecipitation with sequencing
TF	Transcription factor
CUT&Tag	Cleavage under targets and tagmentation
GFP	Green fluorescent protein
mESCs	Mouse embryonic stem cells
WT	Wild-type
FrIP	Fragments in peaks
TSS	Transcription start site
conA	Concanavalin A

Interactions between DNA and proteins are essential for genome organization and regulation. To identify these interactions, it is necessary to precisely map the binding sites of regulatory proteins on chromatin. Chromatin immunoprecipitation with sequencing (ChIP-seq) has long been the gold standard method to map histone modifications and transcription factor (TF) binding sites genome-wide¹. Enzyme-tethering strategies developed recently, such as CUT&RUN², allow mapping of chromatin targets in situ in unfixed cells and provide more biologically relevant profiles of the regulatory landscape while avoiding the risk of epitope masking. The maps obtained with these methods are genome-wide with base-pair resolution and have greatly decreased background. They therefore require significantly lower sequencing depth, which decreases the costs associated with chromatin profiling^{2,3}. However, while CUT&RUN uses lower amounts of input cells than ChIP-seq, it must be followed by DNA end polishing and adapter ligation, which increases the duration of the protocol and prevents its application to single-cell platforms⁴. Newer alternative methods such as Cleavage Under Targets and Tagmentation (CUT&Tag)⁴ and derivatives^{5–12} use a transposome consisting of a fusion protein between the

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IgG-binding protein A and the Tn5 transposase to achieve targeted tagmentation, directly generating fragments ready for PCR and sequencing. CUT&Tag overcomes limitations of ChIP-seq and CUT&RUN, allowing the profiling of chromatin targets with extremely low background in single cells. However, most of these methods require IgG antibodies to bind the desired targets, which is a limitation because specific IgGs are only available for a subset of regulatory proteins¹³. Further, the production of IgGs is costly, lengthy and requires the use of animals.

Recently developed derivatives of CUT&Tag and CUT&RUN that rely on the use of nanobodies overcome some of the limitations of existing techniques. Nanobodies are small, compact antibodies with high antigen-binding affinity and cell penetrability but no cytotoxicity. They can be produced on a large scale and at low cost in bacteria¹⁴. Two techniques that improve multimodal profiling of chromatin targets compared to previous techniques are nanobody-based single-cell CUT&Tag (nano-CT)¹⁰ and nanobody-tethered transposition followed by sequencing (NTT-seq)¹¹. This is done using primary IgG antibodies against different targets from different species combined with fusions between Tn5 and nanobodies that recognize IgG from different species. While these protocols require fewer incubation and wash steps than traditional CUT&Tag, they still rely on the availability of primary IgG antibodies with high specificity from different species for each target.

In contrast, greenCUT&RUN is a method that does not require IgG antibodies. Instead, it uses a protein fusion between MNase and an anti-GFP nanobody. This method allows high-resolution profiling of TFs in cell lines expressing GFP-tagged targets. However, it requires higher input than CUT&Tag (up to 1 million cells depending on the target for greenCUT&RUN compared to 200,000 cells or less for CUT&Tag) and DNA end polishing and ligation steps for library preparation.

Inspired by CUT&Tag and greenCUT&RUN, we developed NanoTag (from “nanobody” and “CUT&Tag”), a technique that replaces traditional IgG antibodies with a fluorophore-binding nanobody to identify DNA-binding sites of fluorescently tagged proteins. NanoTag is a one-step method that uses a fused anti-GFP nanobody-Tn5 protein, which binds directly to tagged proteins. We established NanoTag in mouse embryonic stem cells (mESCs) and used it to profile the histone posttranslational modification (HPTM) H3K4me3 via its protein binder TAF3¹⁵ and the TFs Nanog and CTCF. For this, we used transgenic mESCs expressing GFP-tagged TAF3¹⁶, Nanog¹⁷ or CTCF¹⁸. With NanoTag, we obtained reproducible binding profiles at high resolution and with low background for all targets, that correlated with data obtained by CUT&Tag in mESCs. In sum, NanoTag is a highly versatile, cost-effective and simple method that allows low-input chromatin profiling without using specific IgG antibodies. We anticipate that it will be useful for high-throughput analyses of DNA binding proteins or epigenetic marks that do not have high affinity IgGs using cell lines expressing GFP-tagged targets.

Results

We designed NanoTag to streamline the tagmentation process and thereby provide an advantage to traditional CUT&Tag. With NanoTag, tagmentation is done in a single incubation step while it requires three separate incubations for CUT&Tag (target-specific primary antibody, secondary antibody and pA-Tn5) (Fig. 1a). This is achieved by using cells expressing GFP-tagged target proteins that can be directly bound by the anti-GFP nanobody-Tn5 protein fusion (Fig. 1a). Histone marks such as H3K4me3 can be profiled using cells that express GFP-tagged TAF3, a H3K4me3 reader. Likewise, binding sites of proteins associated with chromatin such as TFs can also be profiled (Fig. 1b). To benchmark NanoTag, we characterized the binding profile of H3K4me3-binding protein TAF3 and the TFs Nanog and CTCF in mESCs expressing GFP-tagged versions of each protein (see Methods). We compared the quality of NanoTag binding profiles to profiles obtained by CUT&Tag in the same cell lines using an anti-GFP antibody. As negative control, we conducted NanoTag and CUT&Tag on wild-type (WT) mESCs not expressing any GFP-tagged target. We chose CUT&Tag to benchmark NanoTag because it is technically the closest even if it is not the most commonly used to profile TFs. Like NanoTag, CUT&Tag can provide high-quality binding profiles with less input material and sequencing reads than methods such as CUT&RUN and ChIP-seq⁴ but is more laborious, time-consuming and expensive than NanoTag. NanoTag and CUT&Tag experiments were performed in duplicate using 400,000 cells per sample and libraries were prepared in parallel (see Methods for NanoTag, CUT&Tag). The distribution of NanoTag and CUT&Tag reads and fragments sizes is similar (Fig. S1a, b) and adaptors are removed by trimming of reads that included them (a minority) (Fig. S1c).

Indirect profiling of H3K4me3 sites in mESCs

When profiling TAF3 by NanoTag (TAF3-NanoTag), we observed enriched coverage at the union of peaks called in the two replicates, supporting the robustness of the peak-calling strategy (Fig. 1c). TAF3-NanoTag peaks were located predominantly at promoters (Fig. 1d), consistent with the enrichment of H3K4me3 at promoters. TAF3-NanoTag peaks were also enriched at genes associated with chromatin organization and histone modification, pathways that are active in mESCs (Fig. 1e). Most (> 80%) of TAF3-NanoTag peaks were captured in both replicates, indicating high reproducibility of NanoTag (Fig. 1f). TAF3-NanoTag coverage showed high reproducibility across replicates and high correlation with both, TAF3-CUT&Tag and H3K4me3-CUT&Tag peak data (91% and 74% respectively) (Fig. 2a). Coverage at peaks for TAF3-NanoTag also highly correlated with H3K4me3 but not with H3K27me3 as expected (Fig. 2a). Importantly, when merging TAF3-NanoTag replicates, NanoTag captured most of the TAF3-CUT&Tag peaks (83%) (Fig. 2b) and most of the TAF3 ChIP-seq peaks (67%) (Fig. S2). Common peaks showed almost identical signal coverage for the two methods (Fig. 2c). We observed that NanoTag coverage is similar to that of CUT&Tag at *Hox* genes cluster, where H3K4me3 signal is expected in mESCs^{19,20} (Fig. 2d). Peaks called in both datasets were associated with genes involved in pathways active in mESCs such as cell cycle phase transition, DNA repair, histone modification and chromatin organization (Fig. 2e). TAF3-NanoTag demonstrated comparable scores for fragments in peaks (FrIP), transcription start site (TSS) and peak enrichment when compared with TAF3-CUT&Tag (Fig. 2f), indicating that NanoTag is

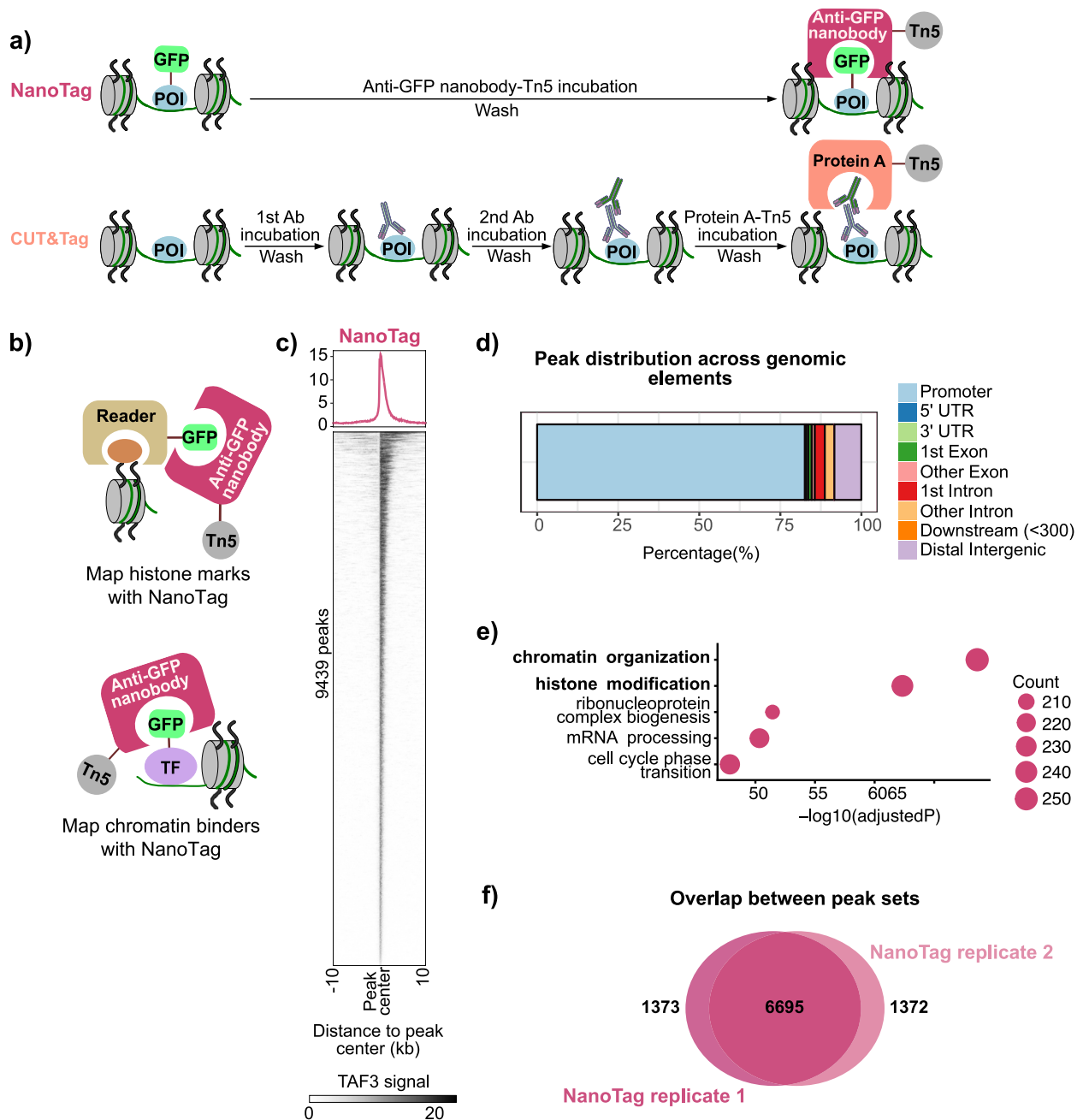


Fig. 1. NanoTag workflow and profiling of TAF3 binding. **a)** Schematic of the steps for NanoTag and CUT&Tag protocols. For NanoTag (magenta, top), permeabilized cells expressing a GFP-tagged protein of interest (POI) are incubated with an anti-GFP nanobody-Tn5 fusion protein. For CUT&Tag (orange, bottom), successive incubation with a primary antibody, a secondary antibody and protein A-Tn5 fusion is required. **b)** Schematic of GFP-tagged chromatin reader protein (top) and chromatin binder protein (bottom) bound to its target histone mark (top) or genomic region (bottom) expressed in mESCs and detected by the anti-GFP nanobody-Tn5 with NanoTag. **c)** Profile plot and heatmap showing TAF3-NanoTag enrichment around the 9439 NanoTag-TAF3 peaks. The profile shown represents the merged coverage of two NanoTag replicates at the union of peaks called in the two replicates. **d)** Summary of TAF3-NanoTag peaks distribution over different genomic elements. The peak set used represents peaks common to both replicates. **e)** Gene ontology (GO) analysis for TAF3-NanoTag peaks. The 5 GO terms with the lowest adjusted p-value are shown with the corresponding gene count. **f)** Overlap between peak sets called in the two TAF3-NanoTag replicates.

comparable in sensitivity to CUT&Tag when profiling TAF3. In contrast, sparse landscapes were produced by both methods in control cells (WT), suggesting comparably low background noise (Fig. 2g). We observed a higher overlap of NanoTag and CUT&Tag TAF3 peaks with promoter regions than for ChIP-seq (Fig. S3), which is consistent with the previously reported bias for Tn5-based methods toward genomic regions with more open

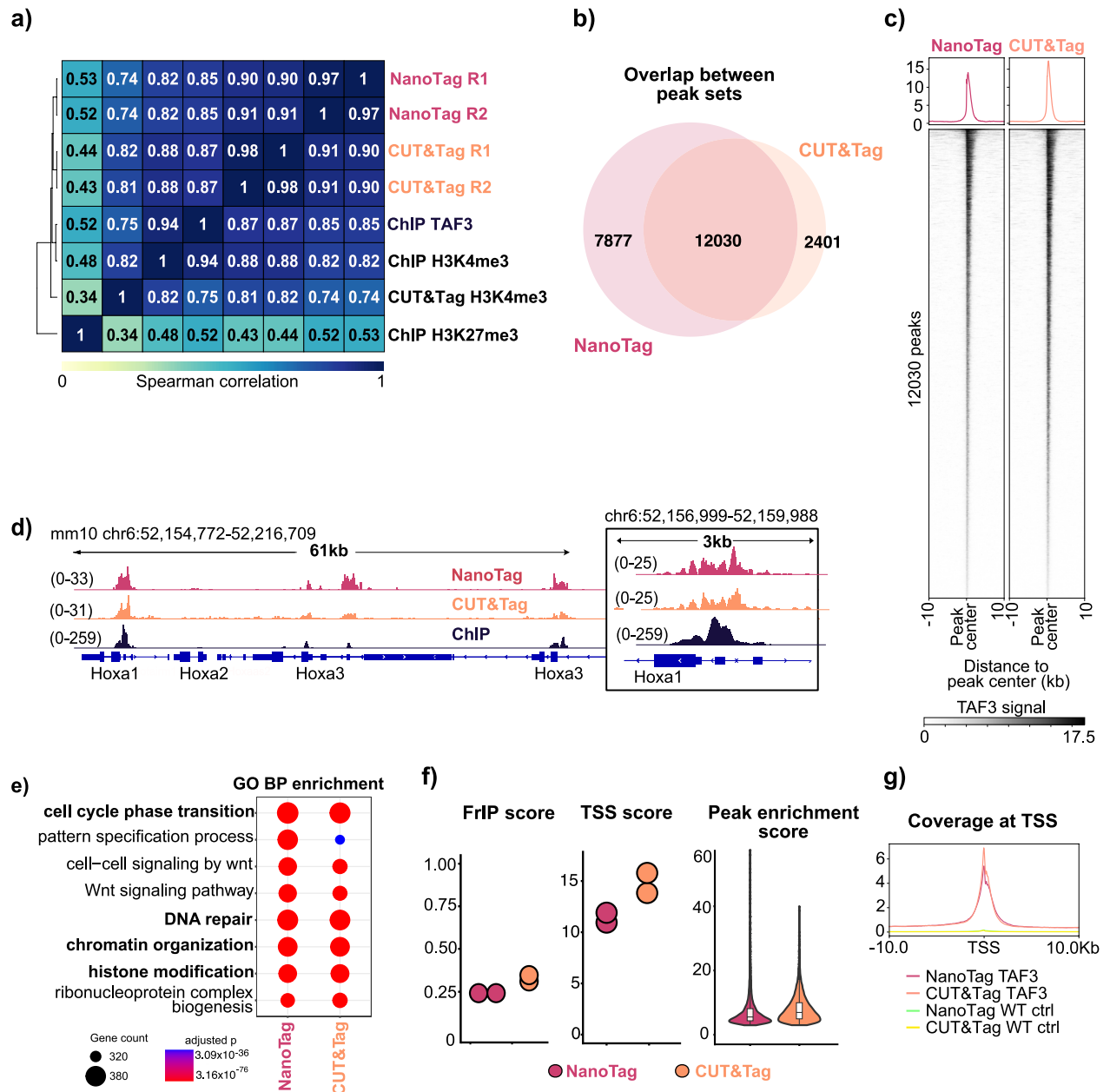


Fig. 2. Comparison of NanoTag and CUT&Tag profiling of TAF3 binding. **a)** Hierarchically clustered correlation matrix of TAF3-NanoTag replicates (magenta, replicate 1 (R1), replicate 2 (R2)), TAF3-CUT&Tag (orange, R1, R2), ChIP-seq targeting TAF3, H3K4me3 and H3K27me3 and CUT&Tag targeting H3K4me3. Spearman correlations were calculated using read coverage at TAF3 peaks common to NanoTag, CUT&Tag and ChIP-seq TAF3 datasets. NanoTag and CUT&Tag datasets were produced in-house, ChIP-seq TAF3 data are from¹⁶, ChIP-seq H3K4me3 and H3K27me3 data are from²², CUT&Tag H3K4me3 data are from²³. **b)** Overlap between NanoTag and CUT&Tag peak sets when profiling TAF3. The peak set corresponding to each method represents the peaks called after merging the two replicates for each method. **c)** Profile plots and heatmaps showing TAF3-NanoTag (magenta) and TAF3-CUT&Tag (orange) enrichment around the 12,030 common peaks. **d)** Genome browser view showing coverage of TAF3 NanoTag, TAF3-CUT&Tag and H3K4me3-ChIP-seq around the *Hoxa* locus. Close-up shows coverage of the same data around *Hoxa1* gene. **e)** Top 14 common GO terms identified as enriched in TAF3-NanoTag and TAF3-CUT&Tag peak sets. **f)** Plots summarizing the FrIP scores (calculated as fraction of reads from each replicate falling into ChIP-seq TAF3 peaks), TSS scores and peak enrichment score distribution for NanoTag and CUT&Tag TAF3 data. FrIP and TSS scores were calculated individually for peak sets called in each replicate. Coverage of individual replicates was merged to calculate the enrichment score at common peaks. **g)** Profile plot comparison of TAF3-NanoTag and CUT&Tag signal around TSS in TAF3-GFP and wild-type (WT) mESCs controls. For each method, the shown profile represents the merged coverage of two replicates.

chromatin²¹. Together, these results demonstrate the suitability of NanoTag to profile H3K4me3 sites in the genome of mESCs using one of its binding partners.

Direct profiling of Nanog and CTCF binding sites in mESCs

Next, we used NanoTag to profile the binding of Nanog and CTCF in mESCs expressing Venus-tagged Nanog¹⁷ and GFP-tagged CTCF¹⁸ (Fig. 3a, Fig. 4a). NanoTag data showed high reproducibility across replicates when profiling Nanog (Fig. 3b) and CTCF (Fig. 4b), high correlation with CUT&Tag and lower correlation with CUT&RUN and ChIP-seq data targeting the respective TF (Fig. 3b, Fig. 4b). Of note, NanoTag data correlate more strongly with ChIP-seq and CUT&RUN data than ChIP-seq and CUT&RUN data correlate with each other. Interestingly, NanoTag shows lower correlation than CUT&Tag with ATAC-seq data when profiling TFs (Fig. 3b, Fig. 4b), consistent with the slightly lower TSS scores when profiling TAF3 compared to CUT&Tag (Fig. 2f). This suggests that NanoTag may have less bias for open chromatin. For Nanog, NanoTag showed higher coverage at peaks common to NanoTag and CUT&Tag (Fig. 3c) and at ENCODE Nanog peaks (Fig. S4a) than CUT&Tag. For CTCF, NanoTag and CUT&Tag coverage was similar at common peaks (Fig. 4c) and at ENCODE

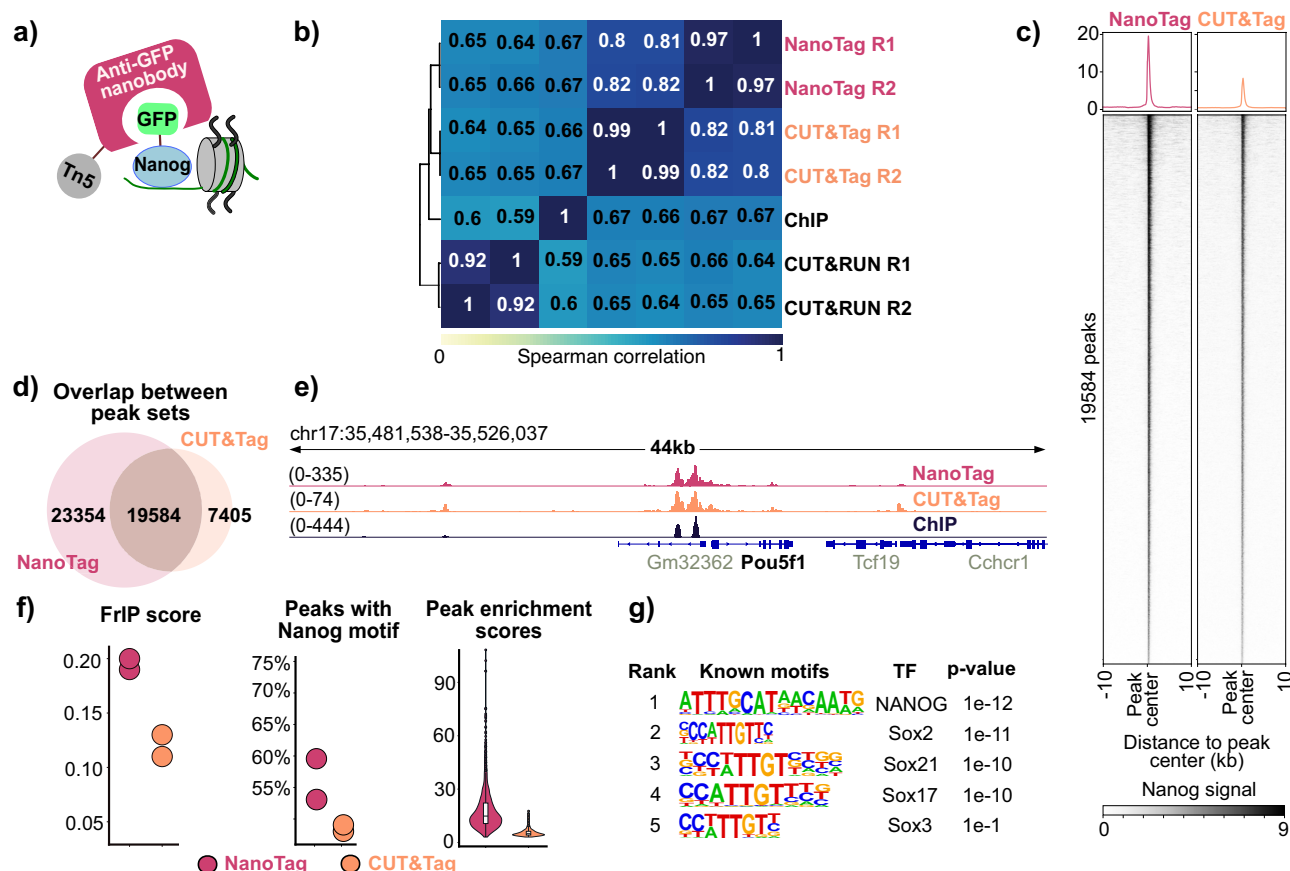
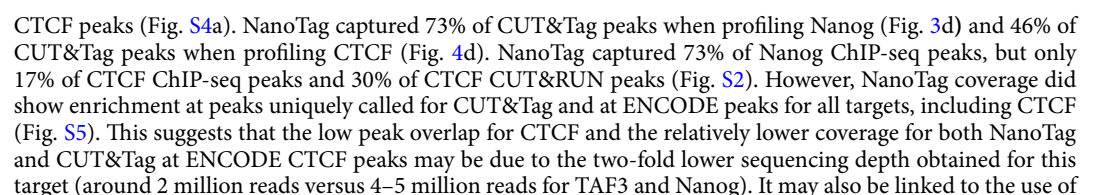


Fig. 3. Comparison of NanoTag and CUT&Tag profiling of Nanog binding. **a)** Schematic of Venus-tagged Nanog expressed in mESCs bound to DNA and detected by the anti-GFP nanobody-Tn5 with NanoTag. **b)** Hierarchically clustered correlation matrix of NanoTag replicates (magenta, R1, R2), CUT&Tag (orange, R1, R2), ChIP-seq and CUT&RUN (R1, R2) data targeting Nanog. Spearman correlations were calculated using read coverage at Nanog peaks commonly called in NanoTag, CUT&Tag and ChIP-seq datasets. NanoTag and CUT&Tag datasets were produced in-house, ChIP-seq data are from²² and CUT&RUN data are from²³ and ATAC-seq data are from²⁵. **c)** Profile plots and heatmaps showing Nanog-NanoTag (magenta) and Nanog-CUT&Tag (orange) enrichment around the 19,584 common peaks. For each method, the signal was merged across two replicates. **d)** Overlap between NanoTag and CUT&Tag peak sets when profiling Nanog data. The peak set corresponding to each method represents the peaks called after merging the two replicates for each method. **e)** Genome browser snapshot showing NanoTag, CUT&Tag and ChIP-seq profiles targeting Nanog around the *Pou5f1* locus. The profile for each method represents the signal obtained from merging two replicates. **f)** Plots summarizing FrIP scores at ENCODE peaks, the proportion of peaks containing Nanog motifs and the enrichment score distribution for common peaks when profiling Nanog using NanoTag (magenta) and CUT&Tag (orange). FrIP scores and proportion of peaks containing the Nanog motif were calculated individually for peak sets called in each replicate. Coverage of individual replicates was merged to calculate the enrichment score at common peaks. **g)** Most highly enriched TF motifs in NanoTag peaks when profiling Nanog. Motifs were identified in the 500 most significant peaks across the two replicates.



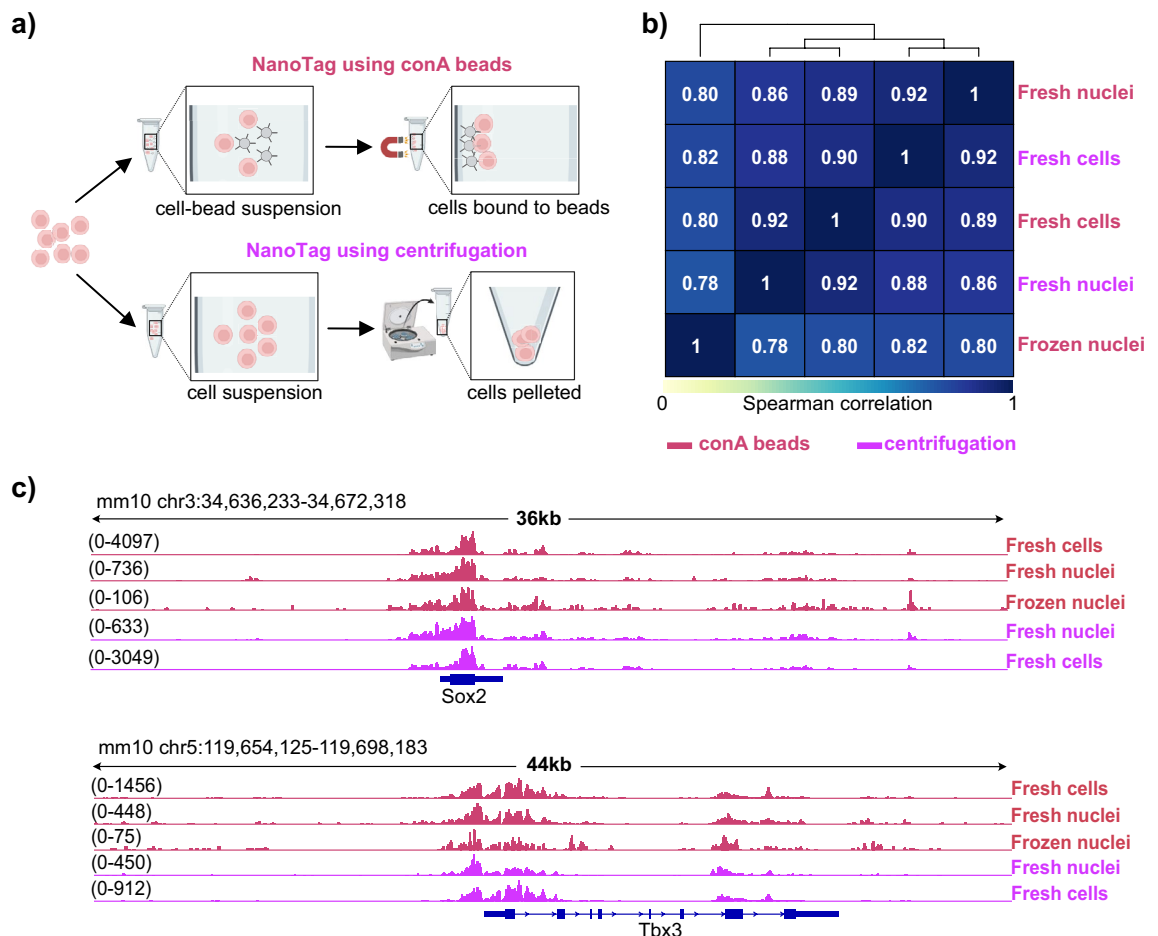


Fig. 5. Variations in NanoTag protocol produce similar results with different input material. **a)** Schematic of two NanoTag protocol variations. NanoTag performed using concanavalin A (conA) magnetic beads to bind cells/nuclei and a magnet to wash the cells bound to beads (top, magenta) and NanoTag performed using centrifugation to pellet cells/nuclei before washing (bottom, purple). **b)** Correlation matrix comparing coverage of NanoTag TAF3 data obtained using different inputs and different protocol variations. Read-level Spearman correlation of TAF3-NanoTag samples prepared using different protocol versions and input material. NanoTag was performed using conA beads as for CUT&Tag (NanoTag conA, magenta) on fresh cells, fresh nuclei or frozen nuclei and using centrifugation (purple, see Methods for details) to pellet cells or nuclei. **c)** Genome browser view showing coverage of TAF3-NanoTag performed using different protocol versions and input materials around the *Sox2* (top) and *Tbx3* (bottom) locus.

unfixed cells unlike for ChIP-seq. CUT&Tag has been previously reported to reproduce only half of ENCODE ChIP-seq peaks for other targets, but these were found to be the strongest peaks²⁴.

In addition, low peak overlap has been reported for various targets across replicates using a single technique e.g. CUT&RUN²⁷. We observed similar signal for both methods at *Pou5f1* locus when profiling Nanog (Fig. 3e) and at *Sox2* superenhancer when profiling CTCF, a locus known to show CTCF binding²⁶ (Fig. 4e). NanoTag showed higher FrIP scores, more peaks containing TF motif and higher peak enrichment scores compared with CUT&Tag when profiling Nanog (Fig. 3f, Fig. S4b) whereas similar scores were found when profiling CTCF (Fig. 4f, Fig. S4b). Further, as expected, NanoTag peaks showed the highest enrichment in Nanog and CTCF motif when profiling the respective TF (Fig. 3g, Fig. 4g). 70% of the top 50 gene ontology (GO) terms overlapped for NanoTag and CUT&Tag when profiling Nanog, and 72% when profiling CTCF. Wnt signaling, a key regulator of mESCs self-renewal, also appeared among the top GO terms in the NanoTag data (fig. S6). These results confirm that NanoTag can be used to directly profile Nanog and CTCF binding sites in mESCs.

Versatility of NanoTag

We tested the applicability of NanoTag to different input material by targeting TAF3 in fresh mESCs and in fresh or frozen nuclei from mESCs (see Methods). Since concanavalin A beads used to bind cells or nuclei are expensive, we also tested a version of NanoTag protocol for which cells/nuclei are centrifuged during collection and washing steps instead of bound by beads (Fig. 5a, see Methods). We obtained highly similar coverage across

protocol versions and inputs (Fig. 5b), suggesting high versatility of NanoTag and no reduction of NanoTag library complexity by freezing nuclei. Libraries showed enrichment at the pluripotency marker *Sox2* locus and *Tbx3* developmental regulator locus across all conditions (Fig. 5c). This was confirmed by qPCR targeting the same genomic regions (Fig. S7). As expected, we observed that libraries produced from cells expressing the GFP-tagged target showed up to 15,000-fold faster amplification (target-dependent) at target regions compared to WT libraries (Fig. S7).

Discussion

We successfully developed NanoTag, a novel method to map binding sites of chromatin-associated proteins and applied it to profile the genomic location of H3K4me3 through its binding partner TAF3 and of Nanog and CTCF in mESCs. Recently, several nanobody-based techniques have been developed including nanobody-based single-cell CUT&Tag (nano-CT)¹⁰ and nanobody-tethered transposition followed by sequencing (NTT-seq)¹¹. These methods use primary antibodies from different species against different chromatin targets, and fusions of secondary nanobodies to Tn5 to allow multimodal profiling of chromatin targets. While powerful, they are not IgG-free and therefore, rely on the availability and specificity of antibodies against the target of interest. Although NanoTag is expected to be suitable for single-cell applications, it requires cell lines that express fusion proteins between the target of interest and a fluorophore, as well as purification of the corresponding nanobody-Tn5 fusions. Nano-CT and NTT-seq may be better suited for multimodal profiling of different targets in individual cells.

GreenCUT&RUN²⁸ uses a fusion between an anti-GFP nanobody and MNase to profile GFP-tagged TFs in situ with high accuracy. This suggests that anti-GFP nanobody can allow reliable genome-wide profiling of GFP-tagged chromatin targets, showing the advantages of an IgG-free approach. But since it is derived from CUT&RUN, it requires high input (up to 1 million cells depending on the target) and includes an extra DNA end polishing and adaptor ligation step. NanoTag combines the benefits of an IgG-free and tagmentation-based method and offers target flexibility by using an anti-GFP nanobody, similar to greenCUT&RUN but with a significantly shorter protocol and allowing lower input. Further, NanoTag is more suitable than greenCUT&RUN to profile histone modifications as Tn5-based tagmentation is preferable to MNase-based CUT&RUN profiling in such cases, similar to CUT&Tag.

While GFP tagging can in some cases disrupt protein function²⁹, tagging was found to not interfere with protein function and genome distribution in each transgenic cell line used in this study^{16–18,28}. This was also the case in other cell lines containing GFP-tagged chromatin binders. While generally concordant for all targets, the genomic profiles obtained by NanoTag, CUT&Tag and ChIP-seq show some differences, which could be explained in several ways. First, cells are crosslinked for ChIP-seq but not for NanoTag and CUT&Tag, which may increase the detection of indirect or weak binding events by ChIP-seq. Further, while for NanoTag, the binding protein and Tn5 are fused, for CUT&Tag the targeting of Tn5 happens through primary and secondary antibody binding. This means that a larger complex of proteins must bind target sites for CUT&Tag profiling, which likely creates steric differences in binding between the two techniques.

NanoTag provides high reproducibility and signal-to-noise ratio and delivers signals similar to CUT&Tag. NanoTag offers a significant advantage over other methods by being IgG-free, making it suitable to profile regulatory proteins that lack high-quality antibodies¹³. It is also a one-step method unlike CUT&Tag which has multiple steps to recruit protein A-Tn5 to desired targets. NanoTag is cost-effective and does not depend on expensive antibodies that require animals to be produced. Instead, it utilizes a nanobody-Tn5 fusion that can be obtained inexpensively and in large quantities using bacteria (animal-free). It is a versatile and efficient method for high-throughput characterization of DNA binding, expected to be an attractive alternative for epigenomic profiling across cells and tissues.

A shortcoming of NanoTag is its high rate of duplicated reads (Table S1). However, we confirmed that this does not alter the validity of the results since only few reads are sufficient to obtain high quality binding profiles. This issue can easily be circumvented by deeper sequencing to obtain enough unique DNA fragments despite high duplication rates. This remains cost-effective since sequencing has become a lot cheaper³⁰. In its current form, NanoTag requires cells that express fluorophore-tagged targets, many of which already exist. Profiling different targets, or the same targets in different cell lines, using NanoTag, requires the use of cell lines that express the desired GFP-tagged target. This can be achieved using CRISPR technologies to generate cell lines stably expressing knock-in GFP fusions³¹. However, this can be laborious and time-consuming as it requires creating and screening single-cell clones. Alternatively, transient transfection or viral transduction with a vector encoding a GFP-tagged target may be faster for exogenous expression and would be sufficient to apply NanoTag.

This study shows that NanoTag using the anti-GFP nanobody can be used to profile GFP-tagged targets, but also targets tagged with similar fluorophores, such as Venus. The anti-GFP nanobody is expected to recognize other GFP derivatives such as YFP and CFP. NanoTag shall be broadly applicable to various chromatin targets including other histone modifications or methylated CpGs. Using mESC lines analogous to the TAF3-GFP line¹⁶, but engineered with GFP-tagged chromatin readers for other marks, NanoTag can be adapted to profile multiple modifications: H3K27me3 with a CBX7 reader, H3K9me3 with a CBX1 reader, and methylated CpGs with MBD1-GFP-expressing cells¹⁶. Multimodal NanoTag could profile targets tagged with different fluorophores using different nanobody-Tn5 fusions. NanoTag can also be envisioned for in vivo applications using fusions between Tn5 and nanobodies or other antibody mimetics that directly bind an endogenous target. Such binders of chromatin-associated proteins already exist^{32–35} and are continuously being developed against new targets. For example, Tn5 could be fused to the anti- γ -H2AX nanobody developed in³⁴ to directly profile endogenous γ -H2AX, a marker for DNA damage.

Conclusions

We developed NanoTag, an IgG-free method to profile DNA–protein interactions and provide evidence that it can be efficiently used to identify the genomic location of a common epigenetic mark and two TFs in mESCs. We benchmarked the method against CUT&Tag and obtained similar quality data. We present a detailed protocol for the method, which can be completed in less time and with lower costs than other techniques.

Methods

Transposome preparation

Using the 3XFlag-pA-Tn5-Fl expression vector⁴ (Addgene plasmid #124601), protein A sequence was replaced with an anti-GFP nanobody sequence that was PCR amplified from the pGEX6P1-GFP-Nanobody vector³⁶ (Addgene plasmid #61838) to add restriction sites for *HindIII* and *EcoRI* flanking the anti-GFP nanobody sequence. The anti-GFP nanobody PCR product and the 3XFlag-pA-Tn5-Fl vector were digested with *HindIII* and *EcoRI* for 1 h at 37 °C. The digested PCR product was purified using the QIAquick Gel Extraction Kit (Qiagen, cat.no. 28115) and the digested vector was purified on a 2% agarose gel. Digestion products were ligated for 1 h at 22 °C then transformed into DH5α competent bacteria. Colonies were picked and sequenced to verify the sequence of the anti-GFP nanobody-Tn5 (pMD1) plasmid (Addgene #215567). The anti-GFP nanobody-Tn5 protein fusion was produced by the Protein Production and Structure Core Facility of the Swiss Federal Institute of Technology (EPFL), Lausanne, Switzerland, following the protocol described by Chen et al. for Tn5 purification³⁷. Briefly, *E.coli* LysY/Iq competent bacteria were transformed with the pMD1 plasmid. One colony was used to inoculate 2L of LB bacterial growth media and overexpression of the anti-GFP nanobody-Tn5 was induced with 0.25 mM IPTG at 18 °C overnight. The bacterial pellet was resuspended in HEGX buffer (NaCl 800 mM, HEPES 20 mM pH 7.5, EDTA 1 mM, Triton X-100 0.1%, Glycerol 10% + protease inhibitor cocktail), sonicated for 2.5 min using 10 s cycles at 70% amplitude and centrifuged for 30 min at 20,000xg. To remove bacterial DNA, the supernatant was treated with 0.275% PEI and centrifuged for 20 min at 20,000xg. The supernatant was bound to chitin beads for 2 h at 4 °C, washed with HEGX buffer and cleavage of the anti-GFP nanobody-Tn5 from the intein tag was induced with 100 mM DTT overnight, eluted and loaded with oligonucleotides for 1 h at 30 °C. Oligonucleotides (synthesized by IDT) were resuspended at 200 μM in water, heated to 95 °C for 5 min and slowly cooled down to hybridize, then mixed and heated to 95 °C for 15 min then cooled down to 50 °C and diluted 1:1 in 2xHEGX buffer with no Triton-X. The protein was dialysed overnight, concentrated, diluted 1:1 in 50 mM HEPES, pH 7.5, 100 mM NaCl, 0.1 mM EDTA, 1 mM DTT, 50% Glycerol, aliquoted, flash-frozen and stored at −80 °C. 2L of bacterial culture yielded 22.5 mL at 23 μM of anti-GFP nanobody-Tn5 transposomes.

Oligonucleotides:

FWD primer for anti-GFP nanobody containing *HindIII* site: GATCAAGCTTAC.

REV primer for anti-GFP nanobody containing *EcoRI* site: CAGAATTCGATC.

Tn5-A: TCGTCGGCAGCGTCAGATGTGTATAAGAGACAG.

Tn5-B: GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAG.

Tn5-rev:/5Phos/CTGTCTCTTATACACATCT.

Cell lines and culture

Wild-type mESCs (HA36CB1, 129×C57BL/6) and mESCs expressing GFP-tagged engineered chromatin readers (TAF3 for H3K4me3) were kindly provided by Tuncay Baubec, Utrecht University and originally reported in¹⁶. R1 mESCs expressing Venus-tagged Nanog were kindly provided by Timm Schroeder, ETH Zurich and originally reported in¹⁷. E14Tg2a AID-eGFP-CTCF, Tir1 mESCs were kindly provided by Elphège Nora, UCSF and originally reported in¹⁸. The TAF3-eGFP construct was integrated once in the mESC genome using recombinase-mediated cassette exchange, leading to stable expression of TAF3-eGFP at levels similar to endogenous chromatin-binding proteins¹⁶. NanogVENUS mESCs are single knock-in clones, which express tagged and untagged Nanog at similar levels¹⁷. eGFP-tagged CTCF mESC are endogenous double knock-in clones¹⁸. The anti-GFP nanobody has been shown to bind with high affinity and specificity GFP, eGFP, YFP and Venus³⁸. Of note, CTCF expression levels are 2–3-fold lower in the AID-eGFP-CTCF compared to wild-type cells^{18,39}. mESCs were cultured on 0.2% gelatin-coated culture dishes in Knockout DMEM supplemented with 15% Fetal Bovine Serum (FBS) qualified for ES cells, 1X GlutaMax, 1X Penicillin/Streptomycin, 1X MEM Non-Essential Amino Acid, 100 μM β-Mercaptoethanol, and 1000 U/ml Leukemia inhibitory factor. Medium was changed daily and mESCs were passaged every 2 days.

NanoTag

A detailed step-by-step protocol can be found on <https://www.protocols.io/view/nanotag-kxygx34zkg8j/v1>. All NanoTag experiments were conducted in duplicate. For all experiments, 400,000 fresh mESCs were collected by treatment with StemPro Accutase at 37 °C for 5 min and centrifugation at 600xg for 3 min and washed with Wash buffer (20 mM HEPES pH 7.5, 150 mM NaCl, 0.5 mM spermidine, protease inhibitors). Concanavalin A (conA) beads were washed with Binding buffer (20 mM HEPES–KOH pH 7.9, 10 mM KCl, 1 mM CaCl₂, 1 mM MnCl₂) as described for CUT&Tag⁴⁰. 10 μL of conA bead slurry was added to the 400,000 cells while gently vortexing and incubated on a rotator for 15 min. Cells bound to conA beads were then resuspended in 100 μL of Dig-wash buffer (0.05% digitonin in Wash buffer) containing 2 mM EDTA and 2 μL of anti-GFP nanobody-Tn5 transposomes and incubated overnight on a nutator at 4 °C. Cells were washed 3 times with 1 mL of Dig-300 buffer (0.01% digitonin in 20 mM HEPES pH 7.5, 300 mM NaCl, 0.5 mM spermidine, protease inhibitors) by incubating on a nutator for 5 min, then resuspended in 300 μL Dig-300 buffer with 10 mM MgCl₂ and incubated at 37 °C for 1 h. Tagmentation was stopped and DNA was extracted using phenol–chloroform

extraction as described for CUT&Tag⁴⁰. When performing NanoTag without the use of conA beads, cells were instead centrifuged at 600xg for 5 min at 4 °C for every resuspension step. When performing NanoTag on nuclei, cells were treated with Nuclei extraction buffer (20 mM HEPES–KOH pH 7.9, 10 mM KCl, 0.5 mM Spermidine, 0.1% Triton X-100, 20% glycerol, protease inhibitors), as described⁴⁰. When performing NanoTag on frozen nuclei, cells were treated with Nuclei extraction buffer and nuclei were frozen as described⁴⁰. Libraries were prepared simultaneously for NanoTag and CUT&Tag samples. To determine the amount of cycles necessary for each library, a 2.5 µL aliquot of the tagged DNA was amplified by qPCR using 2 µL of i5 index, 2 µL of i7 index, 7.5 µL of NEBNext PCR Master mix, 0.15 µL of 100× SybrGreenI dye and 0.85 µL of water with the following program: 72 °C for 5 min, 98 °C for 30 s, and 30 cycles of: 98 °C for 10 s, 63 °C for 30 s and 72 °C for 1 min⁴⁰. 21 µL of tagged DNA was then PCR amplified with the amount of cycles determined by qPCR to yield one third of the maximum fluorescence using 2 µL of i5 index, 2 µL of i7 index and 25 µL of NEBNext PCR Master mix as described for ATAC-seq⁴¹. NanoTag libraries were amplified using 21–22 cycles by PCR (72 °C for 5 min, 98 °C for 30 s, n cycles of 98 °C for 10 s and 63 °C for 10 s, 72 °C for 1 min and held at 8 °C) to achieve a library concentration of at least 2 nM. Libraries were cleaned up as described⁴⁰ using 60 µL of AMPure beads and eluted in 20 µL of 10 mM Tris–HCl pH 8.0. The library profile was verified by TapeStation with the High-sensitivity D1000 kit.

CUT&Tag

CUT&Tag experiments were conducted in duplicate as described⁴⁰ with minor modifications. For all experiments, 400,000 fresh mESCs were collected, washed with Wash buffer and bound to conA beads as described above for NanoTag. Beads were then resuspended in 50 µL Dig-wash buffer (20 mM HEPES pH 7.5, 150 mM NaCl, 0.5 mM spermidine, protease inhibitors, 0.05% digitonin) with 0.1% BSA, 2 mM EDTA and 1:100 anti-GFP antibody (Abcam, cat. no. ab290) or IgG (EpiCypher rabbit IgG, cat. no. 13–0042) and incubated overnight on a nutator at 4 °C. Beads were resuspended in 100 µL of Dig-wash buffer with 1:100 secondary antibody (guinea pig α-rabbit antibody (Antibodies online cat. no. ABIN101961) and incubated on a nutator for 1 h at room temperature. Beads were washed 3 times with 1 mL Dig-wash buffer, resuspended in 100 µL Dig-300 buffer containing 1:250 pA-Tn5 (diagenode, cat. no. C01070001) and incubated on a nutator for 1 h at room temperature. Beads were washed and tagmentation was performed as described above for NanoTag. DNA was extracted with phenol–chloroform–isoamyl alcohol extraction as described in⁴⁰ and stored at –20 °C. CUT&Tag libraries were amplified using 14 cycles and purified as described above for NanoTag.

Sequencing

NanoTag and CUT&Tag libraries were serially diluted to equal molarity and pooled to 400 pM and 10% PhiX was added to the pool. Sequencing was performed on the Illumina NextSeq2000 using 50 bp paired-end sequencing. Standard sequencing conditions for CUT&Tag libraries use 25 bp x 25 bp paired-end reads, which do not include adapter sequences⁴². To account for the longer reads used in this study, the adapter sequences were trimmed before alignment. Since NanoTag libraries show very high duplication rates, sequencing was performed aiming for 25 million read pairs per sample.

Data availability

Datasets generated here were deposited to the Gene Expression Omnibus under accession GSE302691. Public datasets used were: GSE208145³⁹ for CUT&RUN profiling of CTCF in CTCF-GFP-AID mESCs cell line (same cell line used in this study), GSE181104²³ for CUT&RUN profiling Nanog and CUT&Tag profiling H3K4me3 in mESCs, GSE128907¹⁶ for ChIP-seq data profiling TAF3 in mESCs (same cell line used in this study), GSE136488²² for ChIP-seq data profiling Nanog and CTCF in mESCs, GSE136479²² for ChIP-seq data profiling H3K4me3 and H3K27me in mESCs and GSE174024²⁵ for ATAC-seq data in mESCs.

Sequencing data analysis

Data was analyzed as described in⁴². Data obtained from cells expressing the GFP-tagged target was always compared to data from WT cells for both NanoTag and CUT&Tag. Briefly, fastq files were trimmed using trim_galore⁴³ and aligned to the mm10 genome using Bowtie2⁴⁴. Unmapped, unpaired, low-quality, blacklisted, mitochondrial and duplicated reads were filtered using SAMtools⁴⁵. Peaks were called using MACS2 using the replicate files and after merging the two replicate files (q-value 0.1), as in⁴². For all comparisons, the datasets were matched in sequencing read depth by downsampling the CUT&Tag data for each target to the depth of the corresponding NanoTag data after filtering. This was done using SAMtools on the filtered alignment BAM files. Peaks called using downsampled CUT&Tag data were used for all downstream comparisons. Tracks were created using bamCompare with controls (–normalizeUsing BPM –binSize 20). Heatmaps and correlation matrices were created using deeptools. All correlation matrices were generated using the alignment (bam) files. For CUT&Tag data, files were downsampled to match the sequencing depth of the NanoTag files (after filtering, Table S2). For ChIP-seq and CUT&RUN data, downsampling was not performed. Correlations were calculated at peaks common to the NanoTag, CUT&Tag and ChIP-seq datasets for the respective target. To generate coverage tracks for heatmaps, replicate alignment files were merged (and for CUT&Tag downsampled to match the sequencing depth of NanoTag data). Heatmaps were generated using the coverage tracks signal around common peaks for each target. Coverage files were not background normalized, as in⁴². Peaks were annotated using ChIPseeker⁴⁶ and gene ontology (GO) analyses were performed using rGREAT⁴⁷. The most enriched 14 GO terms common between NanoTag and CUT&Tag were displayed in the GO analysis plots. FrIP scores were calculated as the percentage of reads falling into ENCODE ChIP-seq peaks out of all reads for each replicate file when profiling each target. countBam() function in the Rsamtools R package was used to determine the number of total mapped, deduplicated reads of each NanoTag, CUT&Tag and ChIP-seq replicate file. getCounts() function in

the chromVAR R package was used to obtain the count of reads overlapping with ENCODE peaks. FrIP scores were then calculated as the percentage of reads in peaks out of all reads, as recommended by ENCODE and by the CUT&Tag data analysis tutorial by Zheng et al. The script for this calculation can be found <https://github.com/mansuylab/Dimitriu2025/tree/main>. Enrichment scores at peaks called in each replicate file were used as outputted by MACS2⁴⁸ after peak-calling. TSS scores were obtained using the TSSscore function in the ATACseqQC R package⁴⁹. ENCODE peaks were used for calculating FrIP scores and enrichment scores when profiling TFs.

Library enrichment qPCR

To check NanoTag (and CUT&Tag) libraries for enrichment of fragments around desired target loci before sequencing, we devised a qPCR-based strategy. Primers were designed based on ENCODE ChIP-seq data for each target as follows: ENCODE peaks were ranked by enrichment score and two primer pairs were designed for each of the 10 strongest peaks per target. Primers were validated using gDNA extracted from the mESC cell line used for each target. Primers for H3K4me3, Nanog and CTCF loci can be found below. Libraries were diluted sixfold and 2 µL of diluted library was combined with 0.2 µL of 10 µM forward/reverse primer mix, 5 µL of SybrGreen qPCR master mix and 2.8 µL of water. qPCR was conducted using the following program: 95 °C for 5 min and 50 cycles of: 95 °C for 10 s, 60 °C for 10 s, 72 °C for 10 s. To determine the enrichment of NanoTag libraries at target loci, the difference in amplification latency between the libraries from cells expressing the GFP-tagged target and wild-type cells was compared at target loci and normalized to amplification at the gene desert. Enrichment qPCR primers.

Primer name	Target	Sequence
Gene desert_FWD	Control	5'-TCC TCC CCA TCT GTG TCA TC-3'
Gene desert_REV		5'-GGA TCC ATC ACC ATC AAT AAC C-3'
R6_Tbx3_set1_FWD	H3K4me3 (TAF3)	5'-ATG GAT CAG TTA GTG GGG GC-3'
R6_Tbx3_set1_REV		5'-ACG TCT TCT TCG GGT TCC AT-3'
R4_Sox2_set1_FWD	H3K4me3 (TAF3)	5'-ACC TCC GGG ACA TGA TCA GC-3'
R4_Sox2_set1_REV		5'-AGG GGC AGT GTG CCG TTA AT-3'
CTCF-AID_6.2_FWD	CTCF	5'-TAA TGA TAC AGC GAC CCC TGC-3'
CTCF-AID_6.2_REV		5'-GGT TTT GGG CGA CTG CAA AG-3'
CTCF-AID_10.2_FWD	CTCF	5'-GCA GTC TGG AGG GAC ATT CAA CA-3'
CTCF-AID_10.2_REV		5'-CAA ACC AGA AGA TTG GGG GCT-3'
Nanog_r3_set1_FWD	Nanog	5'-CCC CAC AGA CTG AGA ATG GC-3'
Nanog_r3_set1_REV		5'-CCC CTA CCT TTG GCA TCC TG-3'
Nanog_r7_set2_FWD	Nanog	5'-GTG TTA GAG TGC TTT GTC CCC-3'
Nanog_r7_set2_REV		5'-CTC AAA GGA GAC AGG TGC CA-3'

Data availability

A detailed step-by-step NanoTag protocol is available on [protocols.io](https://www.protocols.io). Sequencing data generated for this study were deposited to the European Nucleotide Archive under accession [PRJEB77284](https://www.ebi.ac.uk/ena/record/PRJEB77284). Public datasets used were: [GSE208145](https://www.ebi.ac.uk/ena/record/GSE208145), [GSE181104](https://www.ebi.ac.uk/ena/record/GSE181104), [GSE128907](https://www.ebi.ac.uk/ena/record/GSE128907), [GSE136488](https://www.ebi.ac.uk/ena/record/GSE136488), [GSE136479](https://www.ebi.ac.uk/ena/record/GSE136479).

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Author contributions

MAD developed the original idea, conceived and conducted experiments, analyzed data, produced graphs and figures, interpreted results and wrote the manuscript. LCS and RGAM helped conceive experiments and interpret results in consultation with PLG. IMM contributed to project conception and conduction and data interpretation, provided guidance, corrected the manuscript and provided resources and funding. All authors contributed to the writing of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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